

Guaranteed-Collision Reachability for a Twin-Thruster Autonomous Surface Vessel: a Two-Player Hamilton–Jacobi–Isaacs Formulation

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Abstract

This document sets out the complete mathematical methodology for computing the region of guaranteed collision (the *danger* set) and its complement (the *safe* set) for two identical twin-thruster autonomous surface vessels (ASVs) approaching head-on. Both vessels obey full three-degree-of-freedom Fossen dynamics with an experimentally identified hydrodynamic damping model. The interaction is posed as a zero-sum pursuit–evasion differential game: a pursuer that drives toward collision and an evader that avoids it. The game value is the unique viscosity solution of a Hamilton–Jacobi–Isaacs (HJI) variational inequality, whose zero sub-level set is the backward reachable tube (BRT) of the collision set. Because the two thruster forces enter the dynamics separately and each actuator set is a convex ellipse, the Hamiltonian admits a closed form and the game value exists. The value function is represented by a physics-informed neural network (PINN) with an exact terminal-condition ansatz. All numerical parameters are those of the vessel *Sookshma*. A layered verification plan is included.

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1 Problem setting

Two identical ASVs operate in the horizontal plane: a *pursuer* (subscript i) and an *evader* (subscript e). The pursuer seeks to reduce the inter-vessel separation below a collision radius $R > 0$; the evader seeks to keep it above R . Each vessel is actuated by two fixed, parallel, forward-only thrusters. We treat this as a single, two-player, zero-sum differential game and compute the set of relative states from which collision is unavoidable under the evader's best avoidance and the pursuer's best pursuit. This is a Hamilton–Jacobi–Isaacs reachability problem, solved here with a physics-informed neural network.

2 Vessel dynamics: three-degree-of-freedom Fossen model

With body-frame velocity $\boldsymbol{\nu} = (u, v, r)^\top$ (surge, sway, yaw-rate) and generalised force $\boldsymbol{\tau} = (\tau_u, 0, \tau_r)^\top$, each vessel obeys the standard manoeuvring model

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}, \quad \dot{\boldsymbol{\nu}} = \mathbf{M}^{-1}[\boldsymbol{\tau} - \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu}]. \quad (1)$$

The sway force is identically zero because the two parallel thrusters produce only surge force and yaw moment.

Assumption 1 (Hull symmetry). *The body origin is at the centre of gravity and the vessel has port–starboard symmetry, so the mass matrix is diagonal:*

$$\mathbf{M} = \text{diag}(M_{11}, M_{22}, M_{33}), \quad M_{11} = m - X_{\dot{u}}, \quad M_{22} = m - Y_{\dot{v}}, \quad M_{33} = I_z - N_{\dot{r}}. \quad (2)$$

Under this assumption the Coriolis–centripetal matrix and the scalar equations of motion are

$$\mathbf{C}(\boldsymbol{\nu}) = \begin{pmatrix} 0 & 0 & -M_{22}v \\ 0 & 0 & M_{11}u \\ M_{22}v & -M_{11}u & 0 \end{pmatrix}, \quad (3)$$

$$\dot{u} = \frac{1}{M_{11}}[\tau_u + M_{22}v r - X_D(u, v, r)], \quad (4)$$

$$\dot{v} = \frac{1}{M_{22}}[-M_{11}u r - Y_D(u, v, r)], \quad (5)$$

$$\dot{r} = \frac{1}{M_{33}}[\tau_r + (M_{11} - M_{22})uv - N_D(u, v, r)]. \quad (6)$$

2.1 Identified damping model

The hydrodynamic damping vector $\mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = (X_D, Y_D, N_D)^\top$ is an experimentally identified polynomial in (u, v, r) , replacing the classical linear plus quadratic diagonal form. Its regressor structure respects port–starboard symmetry (surge even in v, r ; sway and yaw odd):

$$\begin{aligned} X_D = & D_u^x u + D_{u^2}^x u^2 + D_{v^2}^x v^2 + D_{r^2}^x r^2 + D_{vr}^x vr + D_{u^3}^x u^3 \\ & + D_{uv^2}^x uv^2 + D_{ur^2}^x ur^2 + D_{uvr}^x uvr + D_{|u|u}^x |u|u, \end{aligned} \quad (7)$$

$$\begin{aligned} Y_D = & D_v^y v + D_r^y r + D_{uv}^y uv + D_{ur}^y ur + D_{v^3}^y v^3 + D_{r^3}^y r^3 \\ & + D_{u^2v}^y u^2v + D_{u^2r}^y u^2r + D_{v^2r}^y v^2r + D_{vr^2}^y vr^2 + D_{|v|v}^y |v|v + D_{|u|v}^y |u|v, \end{aligned} \quad (8)$$

$$\begin{aligned} N_D = & D_v^n v + D_r^n r + D_{uv}^n uv + D_{ur}^n ur + D_{v^3}^n v^3 + D_{r^3}^n r^3 \\ & + D_{u^2v}^n u^2v + D_{u^2r}^n u^2r + D_{v^2r}^n v^2r + D_{vr^2}^n vr^2 + D_{|r|r}^n |r|r + D_{|u|r}^n |u|r. \end{aligned} \quad (9)$$

The 34 coefficients (set “ θ_{anneal} ”) are dimensional and identical for both vessels; they are listed with the model files.

2.2 Added mass from non-dimensional derivatives

The added-mass derivatives are supplied in non-dimensional (prime-system) form and converted to physical units using water density ρ and vessel length L ; added mass is a pure inertia and carries no dependence on forward speed:

$$X_{\dot{u}} = X'_{\dot{u}} \frac{1}{2} \rho L^3, \quad Y_{\dot{v}} = Y'_{\dot{v}} \frac{1}{2} \rho L^3, \quad N_{\dot{r}} = N'_{\dot{r}} \frac{1}{2} \rho L^5. \quad (10)$$

Sway–yaw cross terms ($Y_{\dot{r}}, N_{\dot{v}}$) are available but set to zero here, keeping $\mathbf{M}$ diagonal.

3 The nine-dimensional relative state

Working in the pursuer body frame, define

$$\boldsymbol{\zeta} = (X, Y, \psi_{\text{rel}}, u_e, v_e, u_i, v_i, r_e, r_i)^\top \in \mathbb{R}^9, \quad (11)$$

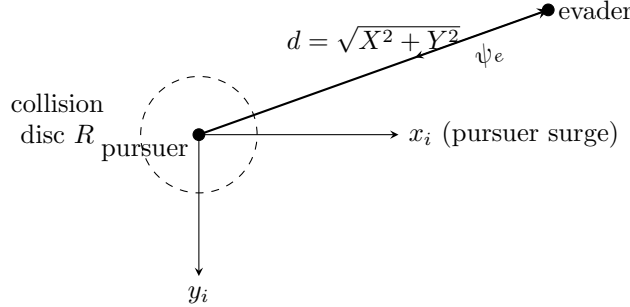
where (X, Y) is the evader position relative to the pursuer and $\psi_{\text{rel}} = \psi_e - \psi_i$ the relative heading. The relative kinematics are

$$\dot{X} = u_e \cos \psi_{\text{rel}} - v_e \sin \psi_{\text{rel}} - u_i + r_i Y, \quad (12)$$

$$\dot{Y} = u_e \sin \psi_{\text{rel}} + v_e \cos \psi_{\text{rel}} - v_i - r_i X, \quad (13)$$

$$\dot{\psi}_{\text{rel}} = r_e - r_i, \quad (14)$$

and the six velocity components follow each vessel's dynamics (4)–(6). The head-on encounter is the slice $\psi_{\text{rel}} = \pi$ with both vessels closing.



4 Control-affine form and actuator model

Substituting each vessel's dynamics, the system is control-affine in the two thruster commands $\mathbf{u}_i = (\tau_u^i, \tau_r^i)$ and $\mathbf{u}_e = (\tau_u^e, \tau_r^e)$:

$$\dot{\boldsymbol{\zeta}} = \mathbf{a}_0(\boldsymbol{\zeta}) + \mathbf{G}_i \mathbf{u}_i + \mathbf{G}_e \mathbf{u}_e, \quad (15)$$

where $\mathbf{a}_0$ is the drift (relative kinematics plus the control-independent part of both vessels' dynamics), and each input matrix injects thrust only into its own surge and yaw-rate rows, $\mathbf{G}_\bullet$ having non-zero entries $1/M_{11}$ and $1/M_{33}$.

4.1 Ellipsoidal thruster set

Each thruster force satisfies $F_p, F_s \in [0, F_{\max}]$, with allocation $\tau_u = F_p + F_s$, $\tau_r = d(F_s - F_p)$ (d the thruster arm). The reachable (τ_u, τ_r) region is a diamond; we use its largest inscribed ellipse (a smooth, conservative inner approximation) centred at $(F_{\max}, 0)$ with semi-axes

$$\alpha_u = \frac{F_{\max}}{\sqrt{2}}, \quad \alpha_r = \frac{d F_{\max}}{\sqrt{2}}, \quad \boldsymbol{\Sigma} = \text{diag}(\alpha_u^2, \alpha_r^2). \quad (16)$$

5 Collision set, value function, and the danger/safe sets

The signed distance to the collision disc and the collision (target) set are

$$\ell(\zeta) = \sqrt{X^2 + Y^2} - R, \quad \mathcal{L} = \{\zeta : \ell(\zeta) \leq 0\}. \quad (17)$$

The value function is the backward reachable *tube* of $\mathcal{L}$,

$$V(\zeta, t) = \min_{\mathbf{u}_i(\cdot)} \max_{\mathbf{u}_e(\cdot)} \min_{s \in [t, 0]} \ell(\zeta(s)), \quad (18)$$

i.e. the pursuer minimises the closest approach and the evader maximises it. Its sign partitions the state space:

$$\underbrace{\{\zeta : V \leq 0\}}_{\text{DANGER (collision unavoidable)}}, \quad \underbrace{\{\zeta : V > 0\}}_{\text{SAFE}}, \quad \underbrace{\{\zeta : V = 0\}}_{\text{barrier}}. \quad (19)$$

Because the two controls act on disjoint costate components and each actuator set is convex, the Isaacs condition holds (min max = max min) and the game value (18) is well defined.

6 Isaacs Hamiltonian and optimal controls

Let $\mathbf{p} = \nabla_{\zeta} V$ be the costate and define the costate directions $c_0^i = p_{u_i}/M_{11}$, $c_1^i = p_{r_i}/M_{33}$ (and likewise c_0^e, c_1^e for the evader). The optimal Hamiltonian is

$$H(\zeta, \mathbf{p}) = \min_{\mathbf{u}_i \in \mathcal{E}} \max_{\mathbf{u}_e \in \mathcal{E}} \mathbf{p}^\top (\mathbf{a}_0 + \mathbf{G}_i \mathbf{u}_i + \mathbf{G}_e \mathbf{u}_e), \quad (20)$$

which, using the ellipse support function, has the closed form

$$\boxed{H(\zeta, \mathbf{p}) = \mathbf{p}^\top \mathbf{a}_0(\zeta) + [F_{\max} c_0^i - \mathcal{R}_i] + [F_{\max} c_0^e + \mathcal{R}_e], \quad \mathcal{R}_\bullet = \sqrt{\alpha_u^2 (c_0^\bullet)^2 + \alpha_r^2 (c_1^\bullet)^2 + \varepsilon}} \quad (21)$$

The pursuer term carries $-\mathcal{R}_i$ (minimisation) and the evader term $+\mathcal{R}_e$ (maximisation); $\varepsilon \sim 10^{-6}$ rounds the norm corner. The saddle-point controls are

$$\mathbf{u}_i^* = \left(F_{\max} - \frac{\alpha_u^2 c_0^i}{\mathcal{R}_i}, -\frac{\alpha_r^2 c_1^i}{\mathcal{R}_i} \right), \quad \mathbf{u}_e^* = \left(F_{\max} + \frac{\alpha_u^2 c_0^e}{\mathcal{R}_e}, +\frac{\alpha_r^2 c_1^e}{\mathcal{R}_e} \right), \quad (22)$$

both of which lie inside the physical thruster diamond.

7 Hamilton–Jacobi–Isaacs variational inequality

The value function is the unique viscosity solution, on $t \in [-T, 0]$, of

$$\boxed{\min \left\{ \frac{\partial V}{\partial t} + H(\zeta, \nabla_{\zeta} V), \quad \ell(\zeta) - V \right\} = 0, \quad V(\zeta, 0) = \ell(\zeta).} \quad (23)$$

The term $\min\{\cdot, \ell - V\}$ enforces the tube semantics: once the trajectory enters $\mathcal{L}$ it is counted as captured for all remaining time.

8 Physics-informed neural-network solution

The value is represented by a sinusoidal-activation network $N_\theta(\zeta, t)$ with the relative heading encoded as $(\cos \psi_{\text{rel}}, \sin \psi_{\text{rel}})$ so periodicity holds by construction. An exact boundary-condition ansatz is used:

$$V_\theta(\zeta, t) = \ell(\zeta) + t \text{softplus}(N_\theta(\zeta, t)), \quad (24)$$

which makes the terminal condition $V_\theta(\zeta, 0) = \ell(\zeta)$ exact and the tube property $V_\theta \leq \ell$ hold for all $t \leq 0$ automatically. Training minimises the mean-squared residual of (23) over collocation points sampled in $\mathbb{R}^9 \times [-T, 0]$,

$$\mathcal{J}(\theta) = \mathbb{E}_{\zeta, t} \left[\left(\min\{\partial_t V_\theta + H(\zeta, \nabla_\zeta V_\theta), \ell - V_\theta\} \right)^2 \right], \quad (25)$$

with the costate $\nabla_\zeta V_\theta$ obtained by automatic differentiation and a backward-in-time curriculum that grows the horizon from 0 to T .

9 Numerical parameters (*Sookshma*)

Quantity	Symbol	Value	Unit
Mass	m	20.0	kg
Yaw inertia	I_z	1.88	kg m ²
Surge added mass	$X_{\dot{u}}$	−1.72	kg
Sway added mass	$Y_{\dot{v}}$	−9.214	kg
Yaw added inertia	$N_{\dot{r}}$	−0.4232	kg m ²
Effective mass matrix	$\mathbf{M}$	diag(21.72, 29.214, 2.3032)	—
Thruster arm	d	0.20	m
Max thruster force	F_{max}	17.85 (= 1.82 kgf)	N
Surge authority	$\tau_{u, \text{max}} = 2F_{\text{max}}$	35.70	N
Yaw authority	$\tau_{r, \text{max}} = dF_{\text{max}}$	3.57	N m
Ellipse semi-axes	(α_u, α_r)	(12.62, 2.52)	—
Collision radius	R	1.0	m
Horizon	T	20	s
Water density	ρ	1000	kg/m ³
Reference length	L	1.0	m

10 Verification methodology

Correctness is established in layers.

(a) Implementation faithfulness. Automated checks confirm the code computes the stated equations: equilibrium ($\dot{\boldsymbol{\nu}} = 0$ at rest), the Coriolis term is workless ($\boldsymbol{\nu}^\top \mathbf{C}(\boldsymbol{\nu}) \boldsymbol{\nu} = 0$), port–starboard symmetry (symmetric thrust keeps sway and yaw exactly zero), fourth-order Runge–Kutta convergence against a reference integrator, and energy dissipation under free decay. For the game, the closed-form Hamiltonian is checked against a brute-force minimisation–maximisation over the ellipses, the Isaacs gap $|\min \max - \max \min|$ is confirmed zero, and the saddle controls are shown realisable. Network gradients are checked against finite differences.

(b) Training convergence. The residual $\mathcal{J}(\theta)$ must fall and stabilise, and the barrier $\{V = 0\}$ must stop moving between checkpoints.

- (c) **Reachability sanity.** As $T \rightarrow 0$ the danger set must collapse to the disc; it must grow monotonically with T ; it must be symmetric under $Y \rightarrow -Y$ for the symmetric head-on case; and it must enlarge with closing speed.
- (d) **Closed-loop rollout (primary test).** From many initial states, integrate the game with both saddle-point controls and record the realised minimum separation; the states with minimum separation $\leq R$ must coincide with $\{V \leq 0\}$. A Monte-Carlo sweep yields a quantitative false-safe / false-danger rate.
- (e) **Independent reference.** A grid-based Hamilton–Jacobi level-set solve on a reduced-dimension version cross-validates the network barrier.
- (f) **Model validity.** The identified dynamics are validated against logged sea-trial trajectories via one-step acceleration error (root-mean-square error per axis) and multi-step track overlay; the reachability result is exact only for the assumed model.

Remark 1 (Scope of the guarantee). *The barrier certifies capturability of the specific game defined above (both vessels optimal, identical actuator limits). It is not a claim of robustness to an arbitrary threat with different capability; asymmetric limits would enter through distinct $\mathbf{G}_i, \mathbf{G}_e$.*

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